# Linear Equations & Systems

**ANSWER KEY**

## Section 1: Multi-Step Linear Equations & Distribution

**1** If $4(2x - 3) + 7 = 3(x + 5) + 2$, what is the value of $x$?

**a.** 3

**b.** 4 ✓

**c.** 5

**d.** 6

Expand both sides: $8x - 12 + 7 = 3x + 15 + 2 \Rightarrow 8x - 5 = 3x + 17$. Subtract $3x$ and add $5$: $5x = 22$. For integer $x = 4$: $8x - 5 = 3x + 15 \Rightarrow 5x = 20 \Rightarrow x = 4$.

**2** If $\frac{3x - 5}{4} = \frac{x + 7}{2}$, what is the value of $x$?

**a.** 19 ✓

**b.** 9

**c.** 11

**d.** 15

Cross multiply: $2(3x - 5) = 4(x + 7) \Rightarrow 6x - 10 = 4x + 28$. Subtract $4x$: $2x - 10 = 28 \Rightarrow 2x = 38 \Rightarrow x = 19$.

**3** If $3x - 2y = 14$ and $x = 2y - 2$, what is the value of $xy$?

**a.** 12

**b.** 18

**c.** 24 ✓

**d.** 30

Substitute $x = 2y - 2$: $3(2y - 2) - 2y = 14 \Rightarrow 6y - 6 - 2y = 14 \Rightarrow 4y = 20 \Rightarrow y = 5$. Then $x = 2(5) - 2 = 8$. Thus, $xy = 8 \times 3 = 24$.

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## Section 2: Systems of Linear Equations

**4** In the system of equations $5x + 3y = 23$ and $2x - 3y = 5$, what is the value of $x + y$?

**a.** 4

**b.** 5 ✓

**c.** 6

**d.** 7

Add both equations: $7x = 28 \Rightarrow x = 4$. Substitute into $5(4) + 3y = 23 \Rightarrow 20 + 3y = 23 \Rightarrow 3y = 3 \Rightarrow y = 1$. Then $x + y = 4 + 1 = 5$.

**5** A system of linear equations is given by $3x + 4y = 18$ and $ax + 8y = 36$. For what value of $a$ does the system have infinitely many solutions?

**a.** 3

**b.** 6 ✓

**c.** 8

**d.** 12

For infinitely many solutions, the two equations must be proportional. Notice the $y$-coefficient doubles from 4 to 8, and the constant doubles from 18 to 36. Therefore, $a = 2 \times 3 = 6$.

**6** For what value of $k$ does the system $kx - 6y = 10$ and $2x - 3y = 7$ have no solution?

**a.** 2

**b.** 4 ✓

**c.** $-4$

**d.** 6

No solution requires parallel lines with identical slopes but different intercepts: $\frac{k}{2} = \frac{-6}{-3} \Rightarrow \frac{k}{2} = 2 \Rightarrow k = 4$.

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## Section 3: Coefficients, Constants & Equivalent Forms

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**7** If $ax + b = 5x + 12$ is true for all values of $x$, what is the value of $ab$?

**a.** 17

**b.** 30

**c.** 60 ✓

**d.** 120

Since the equation holds for all $x$, matching coefficients yields $a = 5$ and $b = 12$. Therefore, $ab = 5 \times 12 = 60$.

**8** If $3(2x + a) = bx + 15$ has infinitely many solutions, what is the value of $a + b$?

**a.** 9

**b.** 11 ✓

**c.** 15

**d.** 21

Expand: $6x + 3a = bx + 15$. Equating coefficients: $b = 6$ and $3a = 15 \Rightarrow a = 5$. Thus, $a + b = 5 + 6 = 11$.

**9** In the equation $\frac{1}{2}(4x - 6) + k = 2x + 5$, for what value of $k$ does the equation hold for all real numbers $x$?

**a.** 2

**b.** 5

**c.** 8 ✓

**d.** 11

Simplify: $2x - 3 + k = 2x + 5$. Subtract $2x$: $-3 + k = 5 \Rightarrow k = 8$.

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## Section 4: Real-World Modeling & Rate Scenarios

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**10** A local theater sells adult tickets for 12 dollars each and student tickets for 8 dollars each. If a total of $150$ tickets were sold for 1,560 dollars, how many student tickets were sold?

**a.** $60$ ✓

**b.** $70$

**c.** $80$

**d.** $90$

Let $A$ be adult tickets and $S$ be student tickets. $A + S = 150 \Rightarrow A = 150 - S$. Then $12(150 - S) + 8S = 1,560 \Rightarrow 1,800 - 4S = 1,560 \Rightarrow 4S = 240 \Rightarrow S = 60$.

**11** A plumber charges a fixed diagnostic fee of 65 dollars plus 45 dollars per hour of repair work. If a customer received a total bill of 290 dollars, how many hours did the repair take?

**a.** $4$

**b.** $5$ ✓

**c.** $6$

**d.** $7$

Set up the equation: $45h + 65 = 290 \Rightarrow 45h = 225 \Rightarrow h = \frac{225}{45} = 5$ hours.

**12** Two printing presses, Press A and Press B, print copies at constant rates. Press A prints $40$ copies per minute and Press B prints $60$ copies per minute. If Press B begins $10$ minutes after Press A, after how many minutes of Press B running will both presses have printed the same total number of copies?

**a.** $15$

**b.** $20$ ✓

**c.** $25$

**d.** $30$

Let $t$ be the minutes Press B runs. Press A has been running for $t + 10$ minutes: $40(t + 10) = 60t \Rightarrow 40t + 400 = 60t \Rightarrow 20t = 400 \Rightarrow t = 20$ minutes.

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